SUMMARY

- First, I read the project instructions that this project is only about **data modeling**, not doing any data analysis.
- I downloaded all the given files and imported them into **Power BI** using the **Get Data** option.
- After loading the files, I opened **Power Query Editor** to clean and prepare the data before building the model.
- In Power Query, I checked all the tables and set the correct **data types**, like whole numbers for IDs, currency for revenue, and date for date columns.
- I removed unnecessary empty or null rows but made sure not to filter out any important data from the fact tables.
- I realized that tables with **Dim** in the name are **dimension tables** (they give details about things), and tables with **Fact** in the name are **fact tables** (they have the transactional data).
- I renamed the tables to make them clear, like **Customer_Dimension, Product_Dimension, Sales_Fact**, so it's easy to understand the model later.
- After finishing all transformations, I closed Power Query and loaded the data into Power BI.
- I also **identified the primary keys (PK) in dimension tables** and **foreign keys (FK) in fact tables**, which helped me know how to connect the tables correctly.
- Then, I went to **Model View** and understood that **Sales_Fact** should be the center of the model since it has the main transactional data.
- I manually created **relationships** between **Sales_Fact** and all dimension tables using **primary key–foreign key logic**.
- I made these relationships **Many-to-One**, so many rows in the fact table link to one row in the dimension table.
- I connected **Sales_Fact** with **Customer_Dimension, Product_Dimension, Region_Dimension, and Date_Dimension**.
- I also linked **Returns_Fact** with **Sales_Fact** using **SalesID** to track returned sales.
- I connected **Returns_Fact** to **Date_Dimension** using **ReturnDateKey**, keeping this relationship **inactive**, as the instructions required.
- I checked all relationships to make sure the **cardinality** and **cross-filtering direction** were correct.
- I understood the **star schema**, with **Sales_Fact in the center** and all dimension tables around it.
- I created **hierarchies** in the dimension tables, like **Date hierarchy, Product hierarchy, and Region hierarchy**, as mentioned in the project.
- I avoided creating any extra tables or relationships that weren't part of the instructions.
- To verify the model, I used only the **Matrix visual**, which is the only allowed visual.
- I created a matrix to show **revenue by product category and region** to check the product and region links.

- Another matrix showed **returns by fiscal year**, which helped check the inactive relationship with **Returns_Fact**.
- I also made a matrix for **revenue by customer segment** to verify the link between **Sales_Fact and Customer_Dimension**.
- I used hierarchy levels in the matrices wherever needed to show the proper hierarchy setup.
- I confirmed that **values changed correctly** when dimensions were changed, which proved the model was working properly.
- Finally, I **saved the Power BI file**, completing the **PR.2 Data Modeler project**.